

George Bibu Final Capstone Assessment

Report

UI-Testing with Playwright

Created a framework structure consisting of 4 main folders: tests, pages, flows, and fixtures to give a proper Page Object Model framework.

The pages folder consists of two LoginPage.ts and BookingPage.ts. These files represent the login page and the search pages of the TripStack application. A folder called flows is created in which TripFlow.ts is created to store the various user flows that can be accomplished within the app. The fixtures folder contains trip.ts so that the tests created use the flows defined in TripFlow.ts. The tests folder is where the test cases are defined.

A single end-to-end test called BusUITest is created in which a user logs in the site, searches for sleeper buses going from Ahmedabad to Delhi, selects the first bus and seat, fills in passenger details and payment details and books the trip successfully.

Secrets such as credentials and details such as phone number, cvv, card number are stored in a .env file at the project root.

Traces for the test are always generated, and videos and screenshots are generated on failure

Q Search tests

All 1Passed 1Failed 0Flaky 0Skipped 0

Capstone

Book a sleeper bus ticket

BusUITest.spec.ts:18

View Trace | 5.6s

chromium

Run

Test Steps

Q Filter steps

> ✓ Before Hooks

1.2s

> ✓ Navigate to "/login" — ./pages/LoginPage.ts:7

551ms

> ✓ Click getByLabel('Email') — ./pages/LoginPage.ts:11

104ms

> ✓ Fill "judy@tripstack.test" getByLabel('Email') — ./pages/LoginPage.ts:12

21ms

> ✓ Click getByLabel('Password') — ./pages/LoginPage.ts:13

31ms

> ✓ Fill "Password@123" getByLabel('Password') — ./pages/LoginPage.ts:14

11ms

> ✓ Click getByRole('button', { name: 'Sign in' }) — ./pages/LoginPage.ts:15

475ms

> ✓ Navigate to "/" — ./pages/BookingPage.ts:7

292ms

> ✓ Click getByRole('tab', { name: 'Buses' }) — ./pages/BookingPage.ts:8

76ms

> ✓ Click getByRole('combobox', { name: 'From' }) — ./pages/BookingPage.ts:9

45ms

> ✓ Click getByRole('option', { name: 'Ahmedabad AMD' }) — ./pages/BookingPage.ts:10

45ms

> ✓ Click getByRole('combobox', { name: 'To' }) — ./pages/BookingPage.ts:11

48ms

> ✓ Click getByRole('option', { name: 'Delhi DEL' }) — ./pages/BookingPage.ts:12

52ms

> ✓ Fill "2026-08-02" getByRole('textbox', { name: 'Date' }) — ./pages/BookingPage.ts:13

13ms

> ✓ Click getByRole('button', { name: 'Search' }) — ./pages/BookingPage.ts:14

200ms

> ✓ Check getByRole('checkbox', { name: 'A/C Sleeper' }) — ./pages/BookingPage.ts:18

85ms

> ✓ Click getByRole('button', { name: 'Select Seats' }).first() — ./pages/BookingPage.ts:19

215ms

> ✓ Click getByRole('button', { name: 'Seat L1 ladies' }) — ./pages/BookingPage.ts:23

62ms

> ✓ Click getByRole('button', { name: 'Continue to passenger details' }) — ./pages/BookingPage.ts:24

184ms

> ✓ Click getByRole('textbox', { name: 'First name' }) — ./pages/BookingPage.ts:28

65ms

> ✓ Fill "Judy" getByRole('textbox', { name: 'First name' }) — ./pages/BookingPage.ts:29

17ms

> ✓ Click getByRole('textbox', { name: 'Last name' }) — ./pages/BookingPage.ts:30

41ms

> ✓ Fill "Joshi" getByRole('textbox', { name: 'Last name' }) — ./pages/BookingPage.ts:31

15ms

> ✓ Click getByRole('spinbutton', { name: 'Age' }) — ./pages/BookingPage.ts:32

50ms

> ✓ Fill "22" getByRole('spinbutton', { name: 'Age' }) — ./pages/BookingPage.ts:33

15ms

> ✓ Click getByRole('textbox', { name: 'Email' }) — ./pages/BookingPage.ts:34

50ms

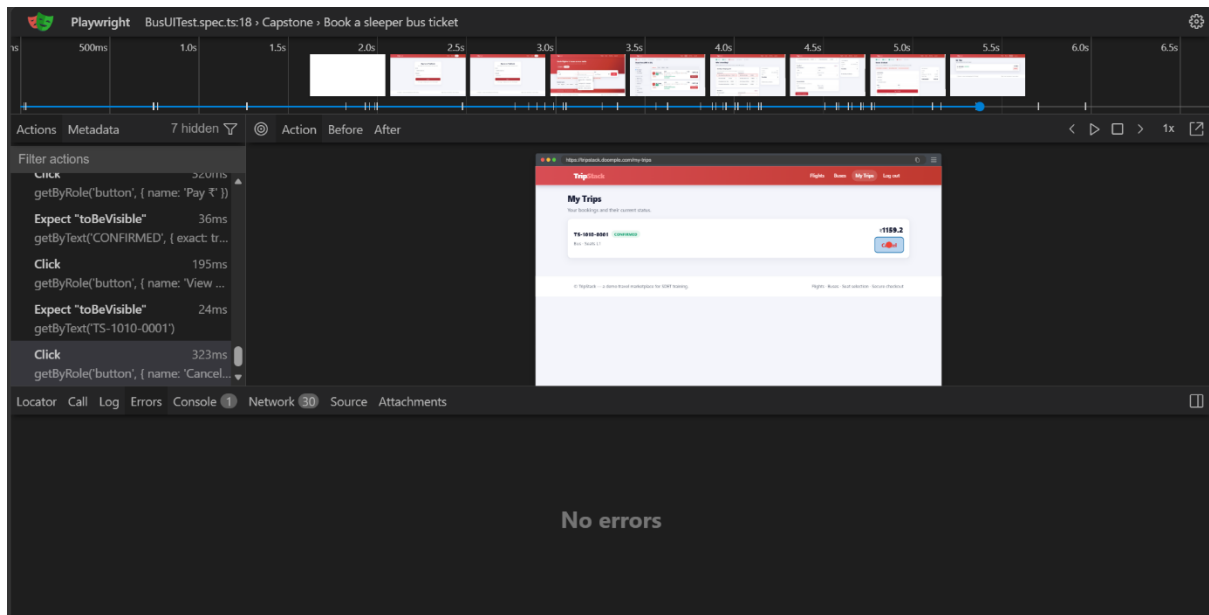
> ✓ Fill "22" getByRole('spinbutton', { name: 'Age' }) — ./pages/BookingPage.ts:33	15ms
> ✓ Click getByRole('textbox', { name: 'Email' }) — ./pages/BookingPage.ts:34	50ms
> ✓ Fill "judy@tripstack.test" getByRole('textbox', { name: 'Email' }) — ./pages/BookingPage.ts:35	17ms
> ✓ Click getByRole('textbox', { name: 'Phone number' }) — ./pages/BookingPage.ts:36	46ms
> ✓ Fill "0000000000" getByRole('textbox', { name: 'Phone number' }) — ./pages/BookingPage.ts:37	14ms
> ✓ Click getByRole('button', { name: 'Continue to payment' }) — ./pages/BookingPage.ts:38	354ms
> ✓ Click getByRole('textbox', { name: 'Name on card' }) — ./pages/BookingPage.ts:42	64ms
> ✓ Fill "Judy Joshi" getByRole('textbox', { name: 'Name on card' }) — ./pages/BookingPage.ts:43	12ms
> ✓ Click getByRole('textbox', { name: 'Card number' }) — ./pages/BookingPage.ts:44	54ms
> ✓ Fill "1234567890123456" getByRole('textbox', { name: 'Card number' }) — ./pages/BookingPage.ts:45	19ms
> ✓ Click getByRole('textbox', { name: 'Expiry' }) — ./pages/BookingPage.ts:46	50ms
> ✓ Fill "05/30" getByRole('textbox', { name: 'Expiry' }) — ./pages/BookingPage.ts:47	11ms
> ✓ Click getByRole('textbox', { name: 'CVV' }) — ./pages/BookingPage.ts:48	47ms
> ✓ Fill "000" getByRole('textbox', { name: 'CVV' }) — ./pages/BookingPage.ts:49	14ms
> ✓ Click getByRole('button', { name: 'Pay ₹' }) — ./pages/BookingPage.ts:50	320ms
> ✓ Expect "toBeVisible" getByText('CONFIRMED', { exact: true }) — ./pages/BookingPage.ts:51	35ms
> ✓ Click getByRole('button', { name: 'View my trips' }) — ./pages/BookingPage.ts:59	196ms
> ✓ Expect "toBeVisible" getByText('TS-1010-0001') — ./pages/BookingPage.ts:60	24ms
> ✓ Click getByRole('button', { name: 'Pay ₹' }) — ./pages/BookingPage.ts:50	320ms
> ✓ Expect "toBeVisible" getByText('CONFIRMED', { exact: true }) — ./pages/BookingPage.ts:51	35ms
> ✓ Click getByRole('button', { name: 'View my trips' }) — ./pages/BookingPage.ts:59	196ms
> ✓ Expect "toBeVisible" getByText('TS-1010-0001') — ./pages/BookingPage.ts:60	24ms
> ✓ Click getByRole('button', { name: 'Cancel' }) — ./pages/BookingPage.ts:64	323ms
> ✓ After Hooks	656ms

Traces



[trace](#)

> Executed in Worker #0



API-Testing with RestAssured

Created a test file called BusAPITest.java to test the full end-to-end flow of booking a bus trip via the provided APIs. The API tests are built and run using Gradle. BusAPITest.java contains two tests, one for the full end-to-end flow and one to test expired tokens.

Command to run test: `./gradlew clean BusAPITest`

Base url: `https://tripstack.doomple.com/api`

APIs tested:

`/auth/login` – POST

`/auth/me` – GET, bearer required

`/buses` – GET

`/buses/{id}/seats` – GET

`/bookings` – POST, bearer required

`/bookings/{id}/pay` – POST, bearer required

`/bookings/{id}/confirm` – POST, bearer required

`/bookings/{pnr}` – GET, bearer required

The second test tests the /auth/me endpoint one more time using an expired token to verify that the appropriate status code (401) is returned.

Allure results are generated in /build/allure-results.

Secrets are stored in a .env file at the root.

```
PS C:\Users\310918\SDet\API> ./gradlew clean BusAPITest
```

```
> Task :BusAPITest
```

```
BusAPITest > expiredTokenSecurityCheck() PASSED
```

```
BusAPITest > BookSleeperBusEndToEnd() PASSED
```

```
BUILD SUCCESSFUL in 9s
```

```
4 actionable tasks: 4 executed
```

▼ build

▼ allure-results

{ } 0d655f89-b835-4d05-a88e-20b962f54f...

{ } 2c736b4e-d537-4ce1-bee1-b01f9ec63e...

{ } 2278fa54-7903-481c-ac5a-88d2b00408...

{ } b3753263-ec91-4449-b87c-3f921bc446...

{ } c22f311a-d271-43b7-8372-c9b044fc0f...